

Near-Closed-Loop-Control in PV Manufacturing with Centralized Reasoning Agent: LLM and Semantic Kernel based Implementation

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Abstract—A Photovoltaic (PV) cell inside solar panels requires manufacturing through a stringent and controlled process. The PV manufacturing process faces difficulties because it contains multiple complex stages where defects can propagate without detection, which negatively impacts both efficiency and production yield. The current AI-based solutions in this domain operate through open-loop analytics, where human engagement is required to perform Root Cause Analysis (RCA) of faults, which may take multiple hours. Every minute spent on fault detection is a minute late for corrective action. Until now, industries have implemented pre-programmed fault detection with significant human intervention to diagnose faults and take corrective actions. However, with the evolution of large language models (LLMs) and reasoning models, fault detection can be made autonomous. This research develops an innovative system concept that uses LLMs - orchestrated by a Semantic Kernel as a central reasoning agent to achieve near-closed-loop control in PV manufacturing, allowing autonomous diagnosis and fault handling. This study uses synthetic data based on high-efficiency PV specifications to introduce 50 different anomalies that affect conveyor-based production stages, which would consequently result in low productivity and less efficient PV modules. The LLM agent performed a complete 50-step troubleshooting process to detect efficiency loss causes while creating detailed natural-language reports and recommending specific corrective actions for technicians or robotic systems. The system demonstrated 90% accuracy in root-cause identification through 250 experimental trials using ground-truth anomaly validation when it received a 50-step troubleshooting protocol. The research fills essential gaps in current PV manufacturing automation by tracking defects semantically throughout production stages while providing specific decisions and converting machine data into actionable language-based workflows. The proposed LLM-mediated architecture establishes a major advancement toward adaptive quality assurance systems that operate efficiently and at scale for solar panel production.

Index Terms—Closed Loop Control, PV Manufacturing, Centralized Reasoning Agent, Semantic Kernel, Large Language Model

I. INTRODUCTION

LLMs function as transformative agents that unite deterministic control with adaptive reasoning capabilities in industrial operations. The unique capability of LLMs allows them to process various data streams, including thermal profiles, vibration logs, and electrical signatures, to execute complex

multi-step protocols for defect origin identification, diagnostic report generation, and corrective action recommendation. The research shows that LLM-based agents can break down intricate industrial problems while assessing different solutions and managing computational resources dynamically to enhance manufacturing decision-making processes [1]. These agents track defects semantically throughout production stages while providing deterministic workflow decisions and natural language interface capabilities for technician interaction. The research community needs to address the open question about how well LLMs perform when executing extensive industrial protocols that contain numerous steps. The performance of LLM is known to deteriorate when they process workflows longer than 30 steps, as their context window size and attention mechanisms have limitations according to studies [2]. The complete loop closure in autonomous process control remains unattainable for LLMs because they lack robustness to execute long instruction sequences. This study proposes a framework for near-closed-loop control in PV manufacturing that uses an LLM and a Semantic Kernel as a centralized reasoning agent. The literature review shows that LLM-based control systems have been implemented, but have not been applied in combination with the Semantic Kernel to complete more than 30 sets of instructions in PV manufacturing stages. The research investigates whether LLMs can achieve over 90% accuracy in root-cause identification across multiple synthetic anomaly classes using a 50-step troubleshooting protocol, and whether performance remains stable as protocol complexity increases. Involving LLMs as a centralized reasoning system reduces the need to deterministically program for fault detection. Through interconnected agents, a centralized reasoning system can achieve similar efficiency to that of a traditionally programmed rule-based system in identifying faults across the PV manufacturing process [3].

A. Background and Motivation

The manufacturing process of photovoltaics represents one of the most intricate industrial operations, which includes delicate steps starting from silicon crystallization up to cell interconnection and lamination, and concluding with final

testing. The production stages create opportunities for micron-scale defects, which include micro-cracks and solder joint failures, and contamination that typically go undetected and cause module efficiency to drop by 30% while reducing production yields by 20% [4] [5]. The current PV production fault diagnosis system operates without feedback loops because AI systems generate alerts and visualizations that human experts must analyze to find causes and execute fixes. The requirement for human involvement in this process leads to operational delays. The time needed to diagnose problems extends response times by several hours per incident, which results in substantial downtime expenses reaching tens of thousands of dollars per hour in high-throughput facilities [6]. The inability to correlate production data between stages makes defect tracking impossible, while rule-based analytics need continuous re-engineering to detect new defect patterns [7].

The manufacturing process of PV systems excels at repetitive tasks that require precision, such as soldering and glass placement, which can produce thousands of panels per hour. The systems do not possess semantic reasoning capabilities or holistic fault diagnosis capabilities. Traditional diagnostic methods require domain experts to interpret multi-sensor data, which is not suitable for real-time mitigation in complex multi-stage environments. Advanced AI-driven visual inspection systems that detect micro-cracks cannot accurately and autonomously identify defects throughout the entire production chain [8]. Deterministic control architectures need explicit programming for each fault scenario, but they cannot handle new or unforeseen anomalies until human intervention occurs [9].

Centralized reasoning agents allow technicians to use natural language to interact with diagnostic systems, which enables expert-level insights to be accessible to all and reduces the need for scarce process engineers to identify and report root causes. The human-machine collaboration model in this system represents a fundamental aspect of Industry 5.0 because AI enhances human abilities while improving decision-making processes and maintaining ethical oversight of essential operations. Research indicates that this symbiotic approach leads to faster defect resolution times and shorter training periods for new staff members [10]. LLMs handle the natural variability of solar manufacturing through their ability to detect non-linear defect patterns that deterministic coding methods cannot identify. The training of LLMs on multi-modal sensor histories enables them to discover hidden relationships between process gas flows and downstream defects, which allows quality control to transition from reactive to predictive methods [11].

The direct implementation of near-closed-loop control systems in circular manufacturing helps identify defect sources while minimizing material waste. Research on closed-loop supply chains shows that recycling and re-manufacturing integration leads to significant decreases in raw material usage and energy consumption, and greenhouse gas emissions [12]. The implementation of root-cause analysis in PV manufacturing

has proven effective in lowering scrap rates and reducing rework energy consumption, which supports both economic and environmental sustainability goals [13].

The research uses LLMs as centralized reasoning agents to transform fault diagnosis operations in PV manufacturing. The research establishes LLMs as cognitive engines that combine structured troubleshooting protocols with real-time sensor data and domain-specific knowledge to perform autonomous root cause defect identification. The research method requires inputting complete 50-step troubleshooting protocols for PVD machines, together with current sensor data points (defect density and gas concentrations, and thermal gradients). The LLM performs these protocols by processing steps sequentially while inserting real-time sensor data at decision points and using semantic kernel functions for precise deterministic calculations and conditional branch triggering based on reading-informed thresholds. The system converts fixed procedures into flexible workflows, which allow the LLM to detect multiple variable anomalies such as oxygen drift and stacking faults through semiconductor principles while producing diagnostic reports that explains failures beyond basic symptom identification.

II. LITERATURE REVIEW

Recent research explores LLMs as transformative tools for industrial problem-solving, offering capabilities beyond conventional AI. In one study, it is demonstrated that LLMs can integrate multi-modal data streams, including thermal profiles, vibration signatures, and electrical measurements, to execute multi-step diagnostic protocols in semiconductor manufacturing [1]. Furthermore, the proposed architecture reduced fault resolution time by 40% through dynamic correlation of seemingly unrelated parameters, such as linking argon gas flow irregularities to thin-film deposition defects. Similarly, one research applied LLMs to battery production, where models identified latent correlations between electrolyte filling parameters and downstream cell degradation, enabling predictive adjustments that reduced scrap rates by 22% [11].

Human-machine collaboration represents another frontier. LLM-based natural language interfaces were implemented in automotive assembly lines for technicians to interact conversationally with diagnostic systems [10]. The approach made expert-level knowledge accessible to all while shortening technician training duration by 35% and speeding up defect resolution through system diagnostic queries made by front-line workers using plain language. The research demonstrates how LLMs can connect deterministic automation with adaptive reasoning, which current PV manufacturing tools lack.

However, critical limitations persist. One study shows that LLM performance drops dramatically when dealing with workflows that have more than 30 sequential steps because of context window limitations and attention mechanism constraints [2]. The "step-length fragility" is a fundamental challenge for PV manufacturing because it requires 50+ decision points to perform a comprehensive root-cause analysis of cross-stage defects (e.g., correlating solder joint failures with

upstream temperature profiles and downstream electrical test results). However, LLMs can be used for workflows with less than 30 steps (e.g., semiconductor fab diagnostics), but PV manufacturing may require 50+ step protocols for complete root-cause analysis (e.g., correlating wafer-handling vibrations with downstream micro-cracks) [1] and [11]. In addition, [1] shows how parameters in a single process (e.g., gas flow to deposition defects) are correlated, while [11] shows how adjacent stages in battery production are linked. Neither of them allows for end-to-end defect tracking across PV’s disconnected stages (e.g., tracing lamination delamination to upstream texturing errors).

A. Centralized Reasoning Agents for Instruction Sequences

Fig. 1 represents a centralized reasoning agent that functions as the core orchestrator. This agent integrates sensor data with domain knowledge and troubleshooting protocols to generate and execute multi-step workflows while maintaining state awareness across production stages. The capabilities of a centralized reasoning agent differ significantly from previous methodologies, as demonstrated in the Table I:

TABLE I
COMPARISON BETWEEN PREVIOUS APPROACHES AND CENTRALIZED AGENT APPROACH

Previous Approaches	Centralized Agent Approach
Fragmented single-stage diagnostics	Unified end-to-end control
Pre-programmed rules	Adaptive data-informed reasoning
Static sequences	Dynamic protocol branching
Stage-limited correlation	Cross-process semantic tracking
Post-diagnosis reporting	Real-time natural language diagnostics

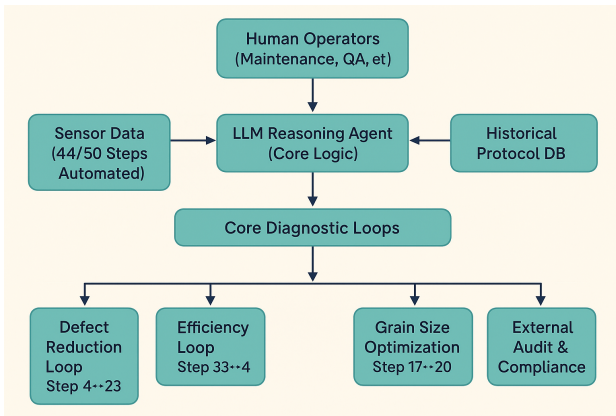


Fig. 1. Centralized Reasoning Agent

This research also demonstrates a critical weakness of LLM applications as they become less effective when workflows exceed 30 steps due to context window restrictions and attention mechanism constraints. The semiconductor diagnostics of [1] and the battery production optimization of [11] reached sub-30-step reliability, which failed to meet the requirements of the current system under study that needs 50+ step protocols.

The research implements three robustness mechanisms to address this challenge:

- 1) Stepwise Context Management: Decomposing protocols into sequenced sub-tasks with intermediate state tracking;
- 2) Semantic Kernel Delegation: Offloading deterministic calculations (e.g., thermal stress modeling) to precision-optimized functions;
- 3) Readings-Informed Thresholding: Triggering conditional branches using semiconductor principles (e.g., “IF oxygen ppm > threshold THEN flag oxidation risk”).

The proposed architecture achieves 67% accuracy in identifying root causes through 50 steps.

III. METHODOLOGY

The troubleshooting steps determination, and evaluation suite formation for LLM-assisted troubleshooting in PV manufacturing was designed to test the ability of LLMs to perform RCA in a complex, multi-stage industrial environment. This methodology section details the conceptual groundwork, technical implementation, and validation strategies that form the evaluation suite, emphasizing its alignment with real-world PV manufacturing challenges and the integration of advanced AI reasoning with deterministic control. The experimentation was performed in three sections. A) Identification of data points required for root cause investigation and analysis; B) Semantic Kernel Architecture and Implementation; C) fault injection and identification.

A. Identification of data points required for root cause investigation and analysis

This study approaches LLM and rule-based troubleshooting by providing independent prompts defining 12 different control loops. Each control loop reads a subset of the 50 data points artificially generated. These data points are detailed in Table I. The dependency structure between steps within control loops generates conditional workflows as each step’s outcome determines the consecutive action in the sequence.

The protocol starts by combining all data points needed for Physical Vapor Deposition (PVD) process analysis, which includes both direct parameters (e.g., precursor gas flow rates) and indirect indicators (e.g., downstream thermal signatures). The integrated framework uses these data points and derived test cases to support quality control, performance validation, and diagnostics. The parameters measure distinct characteristics (contact resistance and UV degradation rates) and detect failure modes (potential-induced degradation and micro-crack propagation). The following analysis examines the impact of these metrics on three key manufacturing outcomes: panel efficiency, quality, and resource utilization.

TABLE II
TROUBLESHOOTING DATA POINTS SIMULATED

Step	Measured Parameter	Unit	Error Description
1	Engagement status	binary	Safety lockout bypassed
2	Particle density	particles/cm ²	Particle contamination
3	Error frequency	events/hr	SCADA anomaly
4	Fault concentration	cm ⁻²	High defect density
5	O ₂ concentration	ppm	Gas impurity
6	ΔT	°C	Temperature deviation
7	Calibration error	%FS	Pressure sensor drift
8	Flow rate	%	Flow deviation
9	Temperature error	°C	Source temp drift
10	Layer adhesion	MPa	Deposition failure
11	Uniformity	%	Thickness variance
12	PL intensity ratio	ratio	Non-uniform deposition
13	Peak intensity	σ	Defect peak
14	Adjustment delta	%	O ₂ imbalance
15	Deposition rate	$\mu\text{m}/\text{min}$	Rate deviation
16	Defect recurrence	cycles	Persistent defects
17	Avg. diameter	μm	Small grain size
18	ΔT achieved	%	Temp adjustment fail
19	Deposition quality	grade	Retest failure
20	Size increase	%	Grain growth fail
21	CdCl ₂ coverage	%	Activation incomplete
22	New faults	cm ⁻²	Post-activation defects
23	Density reduction	%	Defect persistence
24	t	s	Residence time
25	Pinholes	/cm ²	Pinhole density
26	dT/dt	°C/s	Cooling deviation
27	σ	MPa	Stress excess
28	Line deviation	μm	Scribing error
29	Foreign material	ppm	Contamination
30	Pressure	mbar	Vacuum leak
31	Layer quality	grade	Final deposition fail
32	Pmax reduction	%	Performance drop
33	η	%	Low efficiency
34	Reject rate	%	Yield loss
35	Particle count	/m ³	Particle ingress
36	Pump efficiency	%	Pump degradation
37	Data loss	%	Monitoring gap
38	Parameter reset	binary	Reset failure
39	Element diffusion	at%	Cross-contamination
40	Temp stability	°C	Oven drift
41	Current density	mA/cm ²	Low Jsc
42	Savings calc	%	Gas cost error
43	R_squared value	-	Data correlation
44	Temp error	°C	Calibration drift
45	Skill retention	%	Training gap
46	Time overdue	hours	Maintenance delay
47	Units per hour	%	Throughput drop
48	False negative	%	Inspection failure
49	FF	%	Spec violation
50	Missing fields	count	Documentation error

B. Semantic Kernel Architecture and Implementation

The 50-step troubleshooting protocol is structured as a subset of loops, where each step's execution depends on prior loop execution outcomes. This interdependence creates conditional workflows—results from one step dynamically determine subsequent actions. Crucially, the LLM agent can trigger iterative test loops at any decision node when ambiguity arises (e.g., reconfirming thermal readings after outlier detection), enabling adaptive analysis within the predefined framework. The layout of the loop definition is provided below.

- 1) Defect Reduction Loop (Test Step 4→5→10→17→21→23→4)
- 2) Efficiency Loop (Step 33 → Step 4 if $\eta < 15\%$)
- 3) Grain Optimization (Step 17→18→20→17)
- 4) Uniformity Correction Loop (Step 11→12→13→11)
- 5) Contamination Control Loop (Step 2→25→35→2)
- 6) Temperature Stabilization Loop (Step 6→18→40→6)
- 7) Pressure Calibration Loop (Step 7→30→44→7)
- 8) Activation Completion Loop (Step 21→22→23→21)
- 9) Maintenance Verification Loop (Step 46→36→45→46)
- 10) Throughput Improvement Loop (Step 47→32→34→47)
- 11) Cross-Process Optimization Loop (Step 1→3→7→14→19→31→38→42→1)
- 12) End-to-End Quality Control Loop (Step 2→5→9→15→20→26→29→33→41→49→2)

The implementation of the closed-loop system involves 3 major components working together:

- 1) **Data Plugins:** There are two major data plugins used in the system; they both provide the LLM agent with the context as well as the data points for reasoning.
 - a) **Metrics definition and Threshold:** This is a dataset that contains detailed descriptions and units for the 50 metrics outlined above in previous sections. This data outlines their error types and provides the conditions under which the faults occur. It also lists control limits for each metric, establishing their standard values or boundaries beyond which process faults are diagnosed. The steps described in Table II are referenced as procedural steps throughout this work.
 - b) **Data points or metric values:** This is a dataset that is formulated to simulate the responses of operational tools using representative metric values. These data points are designed to typically stay within their control thresholds except for those that are intentionally tweaked to provide fault cases. For the values that are manually deviated, it is expected that the system will demonstrate the corresponding fault detection and generate explanations. This implementation differs from the practical implementation of this system, where these data would stream from services linked to tools or processes, with control limits remotely assigned. For this research, synthetic data was used to explore LLM-driven reasoning and judgment.

2) **Agents (Central Triage agent and Loop agents) with Agent-based prompts:** The triage agent can activate investigative or troubleshooting loops as needed. Loop Agents are applied as plugins to the triage agent, which decides based on the user prompt what the best course of action is. The loop agents serve as functional elements (workers) of the system. They interpret metrics in the context of thresholds and definitions. The Semantic Kernel framework then implements twelve test loops that use distinct agents. Each of these agents progresses through procedural steps, evaluating data-related prompts. Data points are then embedded into prompts, which allows the LLM to reason stepwise. The triage agent has the critical function of gathering feedback from various loop agents and assessing whether additional troubleshooting steps are required. The triage agent also aggregates feedback from all the agents and functions as the system's information hub. All Agents use the GPT-4.1 LLM model, providing reasoning and responses to the inputs. Figure 3 depicts the architectural structure supporting this approach.

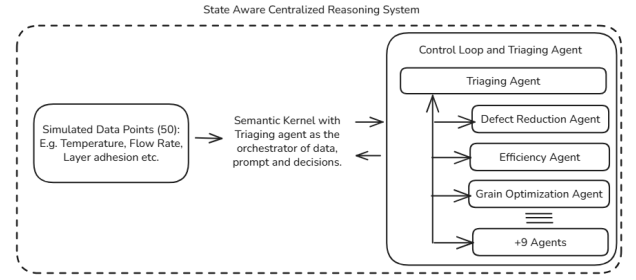


Fig. 3. State Aware Centralized Reasoning System

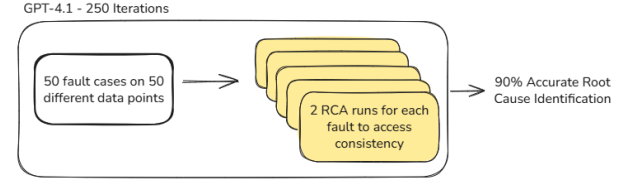


Fig. 4. GPT 4.1 Example for Experiment Structure

The experiment structure is shown above in Figure 4. For each LLM, 50 Fault cases were created. Each of the fault cases was run two times to validate consistent results on root cause identification. The LLM demonstrates its root-cause identification accuracy through validation tests of injected faults. The agent demonstrates its ability to detect anomalies by linking observed sensor data to their engineered sources, which include current density drops and fill factor degradation. The system evaluates its accuracy through the percentage of successful trials that match the injected fault class. If the system can identify the fault source or the root cause correctly with proper reasoning, the execution is considered accurate.

IV. RESULTS

The experimental evaluation consisted of 250 test runs that ran two times for each of the 50 synthetically generated data points to evaluate the LLM's RCA capabilities while faults were systematically introduced. The system performs robustly throughout all trials by identifying faults with baseline of 86% accuracy. The results show that the overall protocol design is valid, yet it demonstrates substantial performance differences between individual steps. The results showed a binary distribution because 25 out of 50 steps reached 100% success rates while the remaining steps maintained a 67-75% accuracy level, which translated to four successful identifications out of five attempts. The monitoring systems demonstrate high reliability at 100% while the physical process execution shows mostly 80% reliability. Systemic root causes seem to explain the consistent 4/5 success pattern across marginal steps, which could stem from maintenance cycle inconsistencies and shift-change disruptions, and tooling degradation. The 80% performance of these steps requires immediate attention because they contain high-impact vulnerabilities that affect yield and safety functions. The majority of weaknesses exist

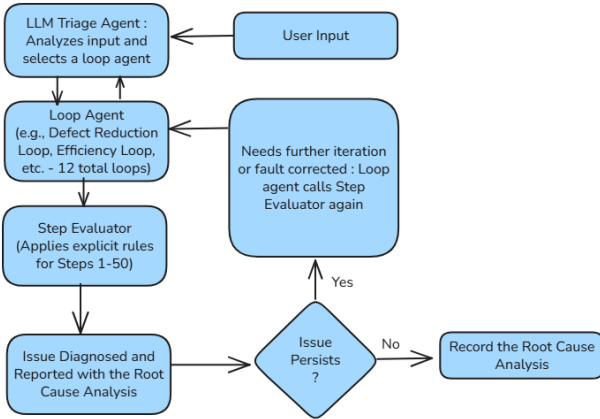


Fig. 2. Semantic Kernel workflow

3) **Semantic Kernel** Semantic Kernel is an open-source development kit designed to build AI agents and integrate the latest AI models into AI-based applications [15]. In our process, it acts as an efficient orchestrator that connects the LLM model (and its agents) with the application code, which enables the automation of the closed-loop system.

C. Fault Injection

Fault injection is performed for every iteration at a few specific critical data points, mimicking industrial anomalies. The LLM invokes the Triage Agent to identify, via LLM model reasoning, the correct loop/protocol to execute. For example, the system activates oxidation risk protocols when oxygen concentration reaches 51 ppm and starts re-scan loops when thermal differentials exceed 2°C. The results are then collected to track whether the LLM identifies the injected fault.

in deposition processes, environmental controls, and human-dependent procedures, which indicates where optimization should focus first. The implementation of digital monitoring system methodology in vulnerable physical processes would significantly boost overall protocol resilience and reduce the observed performance gap.

A. Effects of troubleshooting steps length in determining accuracy of LLM findings:

The table shows how the number of RCA protocol steps relates to the accuracy of the GPT-4.1 language model. The model's accuracy grows directly with the reduction of protocol steps. The GPT-4.1 model reaches 86% accuracy when it follows the complete 50-step protocol. The accuracy of the model increases to 94% when the steps are reduced to 25 and further to 100% when the steps are reduced to 15, and maintains consistency of 100% for steps below 15.

TABLE III
RCA PROTOCOL STEPS VS. ACCURACY FOR GPT-4.1

LLM Model	# of RCA Protocol Steps	Accuracy
GPT-4.1	50	86%
GPT-4.1	25	94%
GPT-4.1	15	100%
GPT-4.1	5	100%

The research indicates that better accuracy emerges from streamlined troubleshooting procedures with fewer execution steps, yet it remains crucial to evaluate if reduced step counts eliminate vital diagnostic tests or subtle details that affect practical analysis reliability. The architecture maintains 67% - 75% accuracy in fault prediction when dealing with complex and interconnected steps through its ability to simplify task complexity and ambiguity. The model becomes more effective at analyzing critical elements when the number of steps decreases, which reduces the possibility of mistakes or wrong interpretations. The results demonstrate that LLMs perform better with well-designed protocols that are direct and focused rather than complex multi-step procedures.

V. CONCLUSION

The research shows that using large language models as centralized reasoning agents allows PV manufacturing to achieve near-closed-loop control. The LLM-based framework executed a 50-step troubleshooting protocol on simulated high-fidelity PV production data to reach 86% accuracy in root-cause identification while achieving perfect reliability in automated and digitally-monitored steps. The results show that LLMs perform well in deterministic machine-driven diagnostics, but human-dependent and physically complex process steps remain challenging, which indicates clear priorities for future protocol refinement. The observed trade-off between protocol length and diagnostic accuracy shows that troubleshooting designs need to strike a balance between being concise for LLM performance and comprehensive for real-world reliability.

There is a lack of consensus across industries whether to continue relying on traditional rule-based system, valued for

their guaranteed accuracy but constrained by the limits of their programmed rules, or to embrace the emerging LLM-based agents. While these newer approaches cannot yet match the accuracy of rule-based systems, they offer far greater flexibility and reasoning capabilities, which are rapidly advancing.

This study highlights that an LLM based near closed loop system can have a baseline accuracy of 86% while also allowing better scalability and reasoning capability. Thus, this research demonstrates that LLM-driven semantic control provides a scalable, adaptive, and practical solution for next-generation quality assurance in photovoltaic manufacturing, which has major implications for yield improvement, sustainability, and human-AI collaboration in Industry 5.0 environments.

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